Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Double Cross #3

Physical address of dairy:

Number and Street	City	County	Zip Code
11071 Youngstown RD	Turlock	Merced	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 07/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X041-X060-X015-XXXX X041-X060-X017-XXXX

B. OPERATORS

Ahlem, Mark

Operator name: Ahlem, Mark

Telephone no.: (209) 668-0807 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95374

This operator is responsible for paying permit fees.

C. OWNERS

Ahlem, Mark

Legal owner name: Ahlem, Mark

Telephone no.: (209) 668-0807 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95374

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	149	0	0	0
Number under roof	402	0	0	0	0	0
Maximum number	402	0	149	0	0	0
Average number	402	0	149	0	0	0
Avg live weight (lbs)	1,150	0	800	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 64 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 11,486.79 *tons per reporting period*

Total nitrogen from manure: 145,261.78 *lbs per reporting period*

After ammonia losses (30% loss applied): 101,683.25 *lbs per reporting period*

Total phosphorus from manure: 24,537.10 *lbs per reporting period*

Total potassium from manure: 75,402.03 *lbs per reporting period*

Total salt from manure: 189,281.70 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 6,331,500 *gallons*

Total nitrogen generated: 11,812.44 *lbs*

Total phosphorus generated: 3,225.48 *lbs*

Total potassium generated: 19,900.56 *lbs*

Total salt generated: 112,086.45 *lbs*

	6,331,500	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	6,331,500	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
DC-5	17	17	2	both	X041-X060-X007-XXXX
DC-6	10	10	2	both	X041-X060-X017-XXXX
Totals for areas that were used for application	27	27	4		
Totals for areas that were not used for application					
Land application area totals	27	27	4		

B. CROPS AND HARVESTS

DC-5

Field name: DC-5

12/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 12/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	282.00 *ton*	As-is		67.8	6,800.00	1,400.00	13,700.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.59	225.60	46.45	454.52	1,143.06

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	446.16 *ton*	As-is		57.3	5,100.00	800.00	5,700.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.24	267.70	41.99	299.19	1,905.10

DC-6

DC-6

Field name: DC-6

12/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 12/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	160.00 *ton*	As-is		67.8	6,800.00	1,400.00	13,700.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	217.60	44.80	438.40	1,102.53

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	234.00 *ton*	As-is		57.3	5,100.00	800.00	5,700.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.40	238.68	37.44	266.76	1,698.61

NUTRIENT BUDGET

A. LAND APPLICATIONS

DC-5 - 12/10/2022: Oats, silage-soft dough

Field name: DC-5

Crop: Oats, silage-soft dough

Plant date: 12/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	107.20	27.20	37.60	1,032.00	170.00 *ton*
Application event totals		107.20	27.20	37.60	1,032.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.40	5.89	91.30	497.17	240,000.00 *gal*
Application event totals		30.40	5.89	91.30	497.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.99	7.36	114.13	621.46	300,000.00 *gal*
Application event totals		37.99	7.36	114.13	621.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.35	6.07	94.16	512.70	247,500.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	212.75	2,063,800.00 *gal*
Application event totals		31.35	6.07	94.16	725.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.68	8.78	81.64	349.90	360,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	270.06	2,619,734.00 *gal*
Application event totals		26.68	8.78	81.64	619.96	

DC-5 - 06/15/2023: Corn, silage

Field name: DC-5

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	282.59	2,741,284.00 *gal*
Application event totals		0.00	0.00	0.00	282.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.56	30.87	122.05	744.57	960,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	248.63	2,411,875.00 *gal*
Application event totals		105.56	30.87	122.05	993.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	219.07	2,125,136.00 *gal*
Application event totals		0.00	0.00	0.00	219.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	212.06	2,057,102.00 *gal*
Application event totals		0.00	0.00	0.00	212.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.47	24.11	95.35	581.70	750,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	206.00	1,998,316.00 *gal*
Application event totals		82.47	24.11	95.35	787.69	

DC-5 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.78	15.43	61.03	372.29	480,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	191.52	1,857,896.00 *gal*
Application event totals		52.78	15.43	61.03	563.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	201.35	1,953,266.00 *gal*
Application event totals		0.00	0.00	0.00	201.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.53	22.38	88.49	539.81	696,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	209.01	2,027,573.00 *gal*
Application event totals		76.53	22.38	88.49	748.83	

DC-6 - 12/10/2022: Oats, silage-soft dough

Field name: DC-6

Crop: Oats, silage-soft dough

Plant date: 12/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	107.20	27.20	37.60	1,032.00	100.00 *ton*
Application event totals		107.20	27.20	37.60	1,032.00	

DC-6 - 12/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.30	6.26	97.01	528.24	150,000.00 *gal*
Application event totals		32.30	6.26	97.01	528.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.21	8.76	135.81	739.53	210,000.00 *gal*
Application event totals		45.21	8.76	135.81	739.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	12.92	2.50	38.80	211.30	60,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	250.48	1,429,300.00 *gal*
Application event totals		12.92	2.50	38.80	461.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.24	9.95	92.53	396.55	240,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	228.19	1,302,100.00 *gal*
Application event totals		30.24	9.95	92.53	624.74	

DC-6 - 06/15/2023: Corn, silage

Field name: DC-6

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

DC-6 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	282.59	1,612,520.00 *gal*
Application event totals		0.00	0.00	0.00	282.59	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	248.63	1,418,750.00 *gal*
Application event totals		0.00	0.00	0.00	248.63	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	123.37	36.08	142.65	870.22	660,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	219.07	1,250,080.00 *gal*
Application event totals		123.37	36.08	142.65	1,089.29	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.69	18.04	71.32	435.11	330,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	212.06	1,210,060.00 *gal*
Application event totals		61.69	18.04	71.32	647.17	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	208.03	1,187,070.00 *gal*
Application event totals		0.00	0.00	0.00	208.03	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	192.46	1,098,216.00 *gal*
Application event totals		0.00	0.00	0.00	192.46	

DC-6 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.29	19.68	77.81	474.66	360,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	198.48	1,132,610.00 *gal*
Application event totals		67.29	19.68	77.81	673.15	

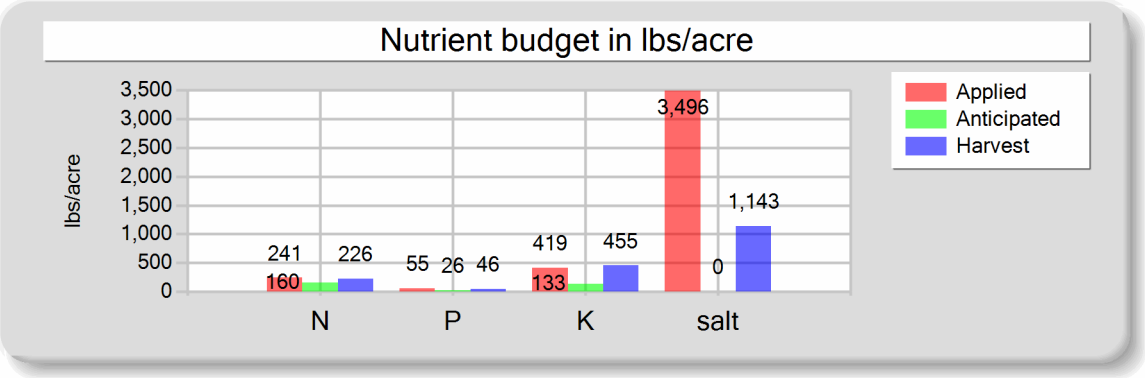
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.84	15.74	62.25	379.73	288,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	204.90	1,169,214.00 *gal*
Application event totals		53.84	15.74	62.25	584.63	

B. NUTRIENT BUDGET

DC-5 - 12/10/2022: Oats, silage-soft dough

Field name: DC-5 Crop: Oats, silage-soft dough Plant date: 12/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	107.20	27.20	37.60	1,032.00
Process wastewater	126.42	28.11	381.24	1,981.23
Fresh water	0.00	0.00	0.00	482.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.62	55.31	418.84	3,496.03
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	225.60	46.45	454.52	1,143.06
Nutrient balance	15.02	8.86	-35.68	2,352.97
Applied to removed ratio	1.07	1.19	0.92	3.06

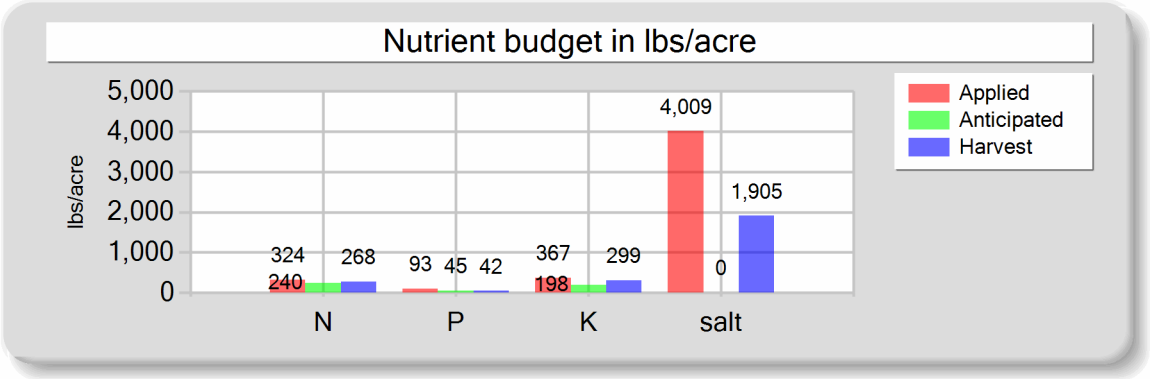
Fresh water applied	
4,683,534.00	*gallons*
172.48	*acre-inches*
10.15	*inches/acre*

Process wastewater applied	
1,147,500.00	*gallons*
42.26	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC-5 - 06/15/2023: Corn, silage

Field name: DC-5 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	317.34	92.79	366.92	2,238.36
Fresh water	0.00	0.00	0.00	1,770.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	324.34	92.79	366.92	4,008.59
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	267.70	41.99	299.19	1,905.10
Nutrient balance	56.64	50.80	67.73	2,103.49
Applied to removed ratio	1.21	2.21	1.23	2.10

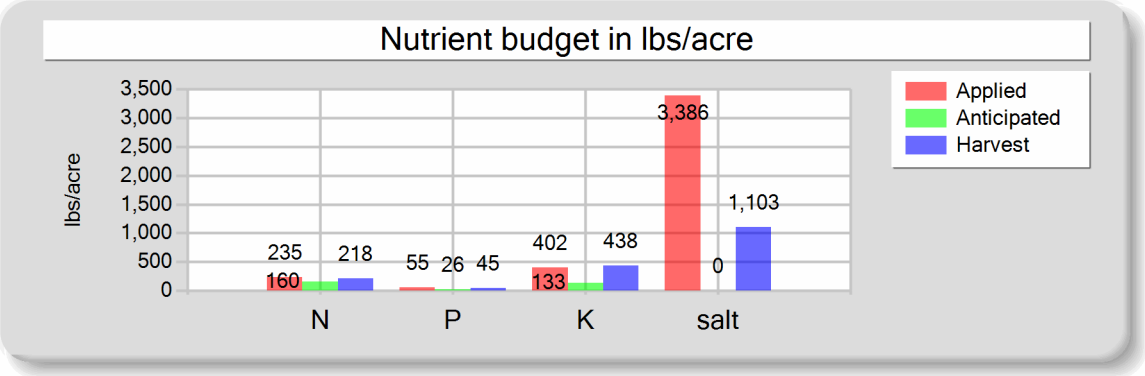
Fresh water applied	
17,172,448.00	*gallons*
632.40	*acre-inches*
37.20	*inches/acre*

Process wastewater applied	
2,886,000.00	*gallons*
106.28	*acre-inches*
6.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC-6 - 12/10/2022: Oats, silage-soft dough

Field name: DC-6 Crop: Oats, silage-soft dough Plant date: 12/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	107.20	27.20	37.60	1,032.00
Process wastewater	120.67	27.48	364.16	1,875.62
Fresh water	0.00	0.00	0.00	478.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.87	54.68	401.76	3,386.29
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	217.60	44.80	438.40	1,102.53
Nutrient balance	17.27	9.88	-36.64	2,283.76
Applied to removed ratio	1.08	1.22	0.92	3.07

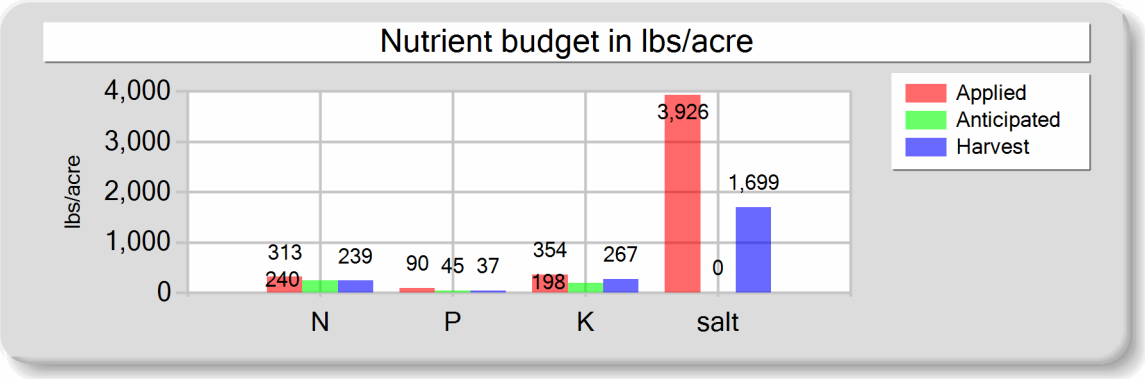
Fresh water applied	
2,731,400.00	*gallons*
100.59	*acre-inches*
10.06	*inches/acre*

Process wastewater applied	
660,000.00	*gallons*
24.31	*acre-inches*
2.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

DC-6 - 06/15/2023: Corn, silage

Field name: DC-6 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	306.19	89.53	354.03	2,159.72
Fresh water	0.00	0.00	0.00	1,766.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	313.19	89.53	354.03	3,925.93
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	238.68	37.44	266.76	1,698.61
Nutrient balance	74.51	52.09	87.27	2,227.32
Applied to removed ratio	1.31	2.39	1.33	2.31

Fresh water applied	
10,078,520.00	*gallons*
371.16	*acre-inches*
37.12	*inches/acre*

Process wastewater applied	
1,638,000.00	*gallons*
60.32	*acre-inches*
6.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Separated Solids

Sample and source description: 2022 Fall Separated Solids

Sample date: 11/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,800.00	6,800.00	9,400.00	1,530.00	6,000.00	1,700.00	6,000.00	970.00		25.80
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,900.00	2,600.00	2,300.00							10.20
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	4,300.00	14,200.00	12,300.00	5,600.00	3,100.00	6,400.00	887.00		65.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	703.00	412.00	0.00	0.00	69.40	902.00								6,740.00	3,120
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	258.00	179.00	0.00	0.00	50.00	775.00								5,190.00	4,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	151.00	104.00	0.00	0.00	49.70	462.00								2,820.00	1,980
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	224.00	120.00	0.00	0.00	65.50	259.00	79.40	56.30	62.90	1,100.00	0.00	89.60	119.00	2,640.00	1,580
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	361.00	277.00	0.00	0.00	23.90	428.00								4,500.00	4,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

T.I.D Canal

Canal

Sample description: Canal

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	210
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

DC-5 - 12/10/2022: Oats, silage-soft dough

DC-5 - 12/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,800.00	1,400.00	13,700.00		10.70
DL	100.00	100.00	100.00		0.01

DC-5 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	800.00	5,700.00		8.50
DL	100.00	100.00	100.00		0.01

DC-6 - 12/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,800.00	1,400.00	13,700.00		10.70
DL	100.00	100.00	100.00		0.01

DC-6 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	800.00	5,700.00		8.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

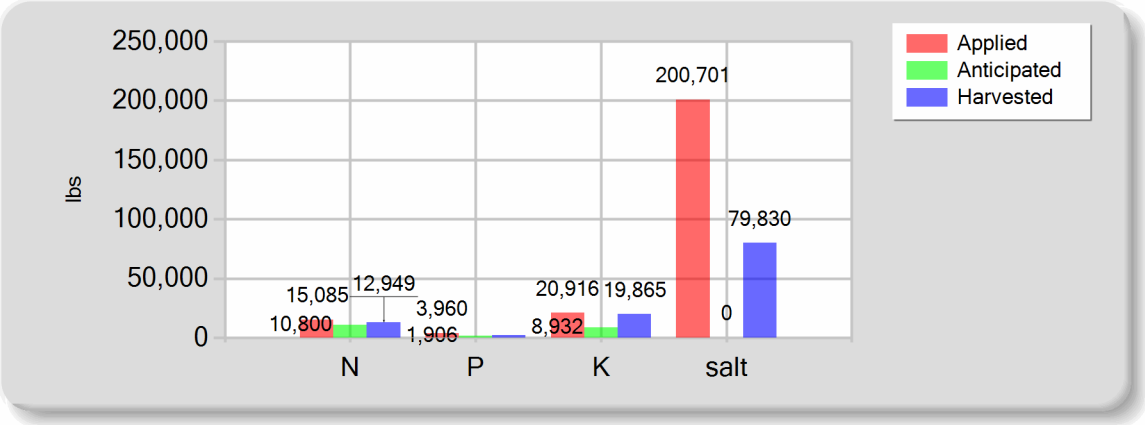
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

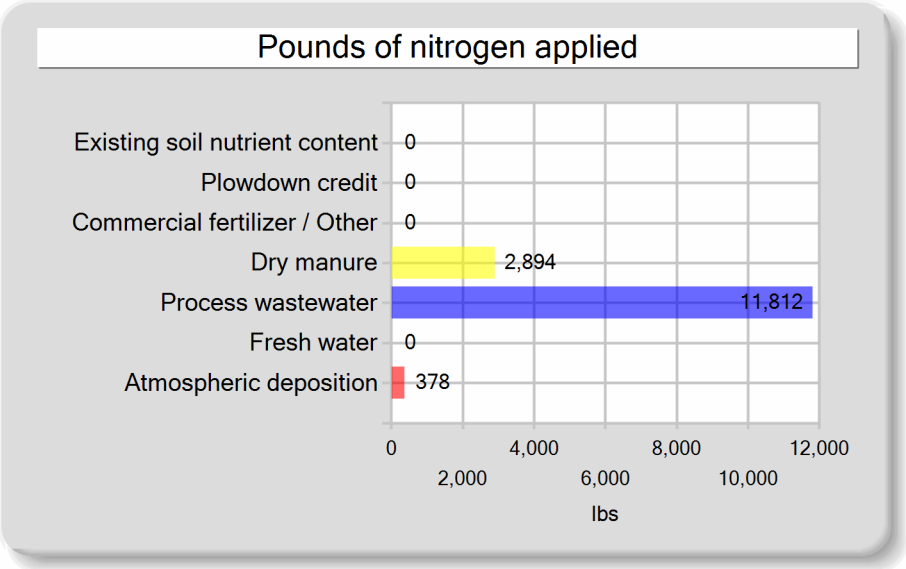
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

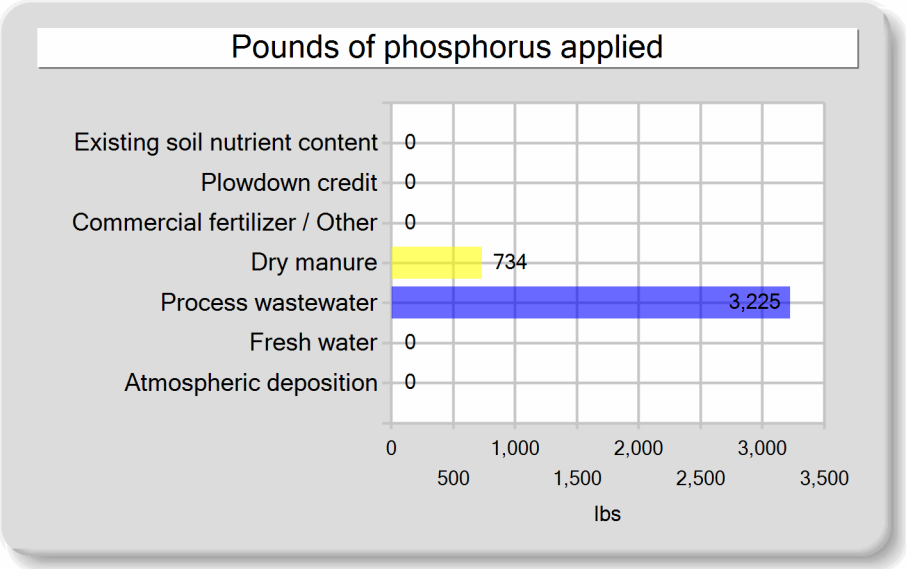
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	2,894.40	734.40	1,015.20	27,864.00
Process wastewater	11,812.44	3,225.48	19,900.56	112,086.45
Fresh water	0.00	0.00	0.00	60,750.26
Atmospheric deposition	378.00	0.00	0.00	0.00
Total nutrients applied	15,084.84	3,959.88	20,915.76	200,700.71
Anticipated crop nutrient removal	10,800.00	1,906.20	8,931.60	0.00
Actual crop nutrient removal	12,948.83	2,325.86	19,864.62	79,830.15
Nutrient balance	2,136.01	1,634.03	1,051.14	120,870.56
Applied to removed ratio	1.16	1.70	1.05	2.51

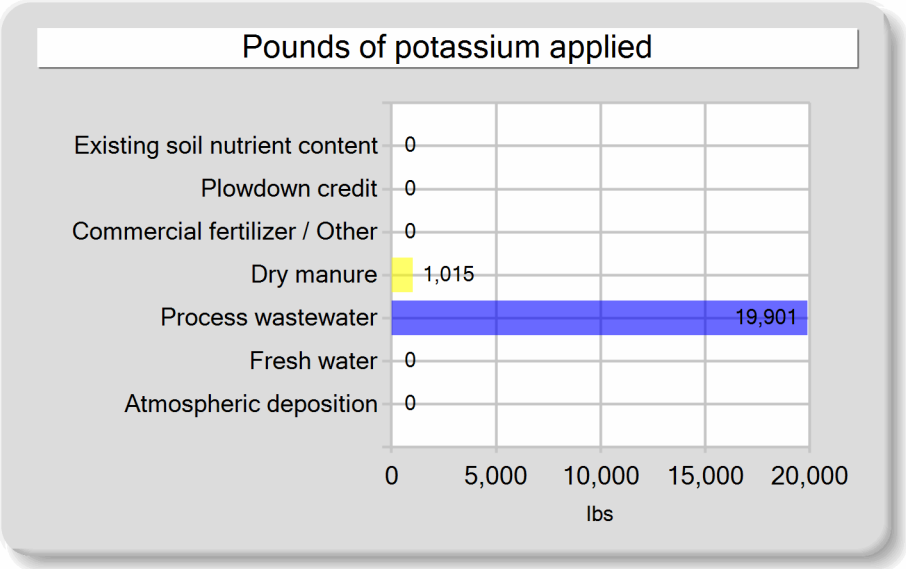
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

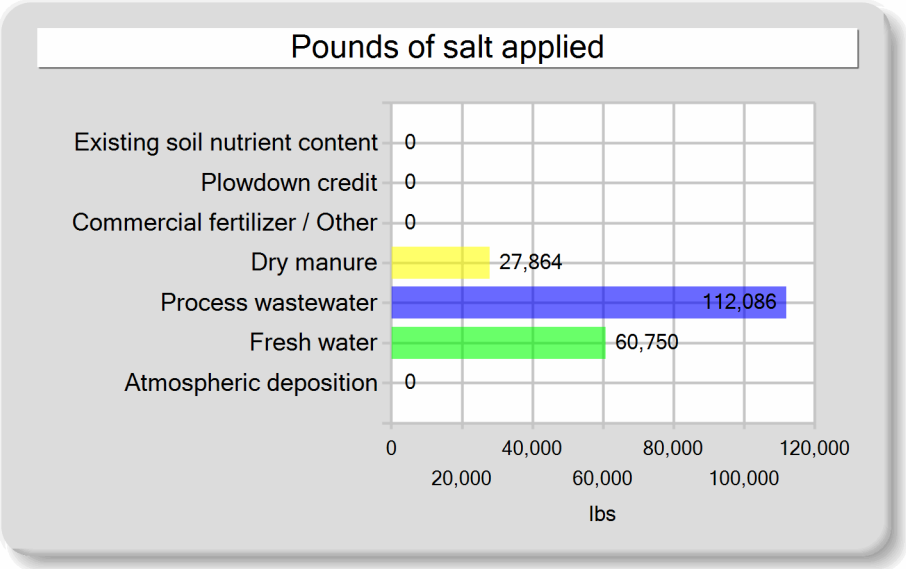


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mark Ahlem Mark Ahlem (May 21, 2024 14:07 PDT)	Mark Ahlem Mark Ahlem (May 21, 2024 14:07 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Mark Ahlem	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 21, 2024	May 21, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Double Cross #3
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
House DW	T3291012A	Electrical Conductivity (EC)	EPA 120.1 and	0.383	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.25	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	245	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)	1.20
Electrical Conductivity (EC)	97.6			0.093
Nitrate Nitrogen (NO3-N)	96.9			2.49

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Double Cross #3
23546 W. American Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn	T3291012B	Electrical Conductivity (EC)	EPA 120.1 and	0.386	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.17	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	247	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)	1.20
Electrical Conductivity (EC)	97.6			0.093
Nitrate Nitrogen (NO3-N)	96.9			2.49

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: DC#3
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: SB
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	10/10	4:15	Domestic	House	DW	16.0c	✓				
2	↓	4:30	well	milk barn		15.6'	✓				
3			✓								
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st		FRAG				
2nd		DENELE ANALYTICAL, INC.	10-18-23 9:45			T3291012

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

2023 Annual Report Double Cross #3

Final Audit Report 2024-05-21

Created:	2024-05-21
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAA3g5E5inD6iAN5jNBKXmRRwBg8a-WoDrC

"2023 Annual Report Double Cross #3" History

Document created by Kristen Jones (kjones@fragservices.com)
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